# STAFF REVIEW · H446 028 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

## Do Now — source answers

1. 11110011 . Start with +13 = 00001101. Invert the bits → 11110010, then add 1 → 11110011.

2. Q = ¬A ∧ ¬B . NOT of an OR becomes AND of the individually negated inputs. Both expressions are equivalent.

3. [4, 6, 2, 8, 9] . Compare neighbouring items and swap where needed: [4,9,6,2,8] → [4,6,9,2,8] → [4,6,2,9,8] → [4,6,2,8,9].

4. $0.110100 \times 2^2$ . Shift the mantissa point two places right (0.001101 → 0.110100), so decrease the exponent from 4 to 2. The numerical value stays the same.

## I Do — source worked steps

1. Read the input combinations — Only the two bottom cells contain 1. They represent A=1, B=0 and A=1, B=1 .

2. Circle the adjacent 1s — Both outputs are 1 and the squares share an edge.

3. Compare each input

4. Write the simplified expression — Keep A because it remains 1. Remove B because it changes between 0 and 1. — Check: Q is 1 for both values of B whenever A = 1.

## We Do — guided CHECK

1. 2. Verify the grouping
They form one vertical group of two.
3. What stays fixed?
For the circled cells, write A and B as 0 or 1. Which value changes?
4. Keep or remove
A: 0 → 1 (changes); B: 1 → 1 (always 1).
5. Write Q
Which input remains after removing the one that changes?
6. Explain the result

2. 1. Identify the group
Circle both 1s in one horizontal pair.
2. Verify the grouping
The top row contains the pair.
3. What stays fixed?
State whether A and B are always 0, always 1, or changing.
4. Keep or remove
A: 0 → 0 (always 0); B: 0 → 1 (changes).
5. Write Q
What expression represents A when A remains 0?
6. Explain the result

3. 1. Identify the group
Circle the largest useful pairs; one cell may belong to both.
2. Verify the grouping
Bottom row and right column: two groups sharing one 1.
3. What stays fixed?
For each pair, which input remains fixed at 1?
4. Keep or remove
Bottom row: A=1, B changes → A. Right column: B=1, A changes → B.
5. Write Q
Write both terms, then combine with OR.
6. Explain the result

4. 1. Identify the group
Write the BC columns in Gray-code order: 00, __, 11, __. Then circle the four 1s.
2. Verify the grouping
Column order is 00, 01, 11, 10. All four 1s form one 2×2 group.
3. What stays fixed?
Across the four cells, which of A, B and C changes?
4. Keep or remove
A: 0 ↔ 1 (changes); B: 0 ↔ 1 (changes); C: always 1.
5. Write Q
Keep the input that stays 1 and remove both changing inputs.
6. Explain the result

## You Do — source mark schemes

1. Circle: Both 1s in the top row make one pair.
Compare: A=0 in both cells. B changes from 0 to 1.
Simplify: A stays 0, so use NOT A (¬A). B changes, so remove B.
Check: Q is 1 whenever A is 0.

2. Circle: A single 1 at A=1, B=1 cannot join another output 1. A group of one is valid.
Compare: A and B are both fixed at 1.
Simplify: Keep both positive terms with AND.
Check: Q=1 only for A=1 and B=1.

3. Group 1: The top row has A=0 throughout; B changes. The term is ¬A.
Group 2: The right column has B=1 throughout; A changes. The term is B.
Overlap: The top-right 1 belongs to both groups; this is allowed.
Simplify: Combine the two terms with OR, giving ¬A ∨ B.

4. Ordinary binary: 00 → 01 → 10 → 11
Gray code: 00 → 01 → 11 → 10
Each neighbouring Gray-code label differs in exactly one bit.
Complete the labels: 00, 01 , 11, 10 .
Why 01 and 11 are neighbours: Only the first input bit changes (0 → 1); the second bit stays 1.
Why ordinary binary order is unsuitable: 01 → 10 changes both input bits, so those columns are not adjacent in a K-map.
Edge connection: 10 ↔ 00 also changes only one bit; the left and right edges touch conceptually.

5. Circle: All four 1s form a 2×2 group across BC=01 and BC=11 and both A rows.
Compare: A changes 0 ↔ 1; B changes 0 ↔ 1; C remains 1.
Simplify: Remove A and B. Keep C.
Check: Q=1 whenever C=1, whatever A and B.

6. Circle: The two 1s are in row A=1, under BC=01 and BC=11.
Compare: A stays 1; C stays 1; B changes 0 → 1.
Simplify: Keep A and C joined by AND. Remove B.

7. Group 1: In row E=1, columns FG=01 and 11 give E ∧ G. F changes while E and G stay 1.
Group 2: In row E=1, columns FG=11 and 10 give E ∧ F. G changes while E and F stay 1.
Overlap: FG=11 is in both pairs and that is allowed.
Simplify: Join the two AND terms with OR. An equivalent factored form is E ∧ (F ∨ G).
Adapted classroom K-map based on the OCR Boolean Algebra popcorn scenario; not a reproduced OCR K-map.

## Exit Ticket — source answer

1. True. A variable that changes within a valid K-map group is eliminated from the simplified term.